

Alan J. Kay

SENIOR SOFTWARE ENGINEER II · MINNEAPOLIS, MN

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SUMMARY

Senior software engineer with 11+ years in platform and product engineering, delivering cloud foundations, developer experience, and AI enablement at enterprise scale. I take projects from idea to adopted — building the solution, enabling the people who use it, and measuring the results. Recent impact includes company-wide AI tool rollouts (GitHub Copilot, Claude Code), cloud platform foundations supporting 3,500+ AWS accounts, and developer-productivity programs with \$10M+ annual ROI.

PROFESSIONAL EXPERIENCE

Principal Financial Group

Des Moines, IA

Sr. Software Engineer II · IS ExE PDIO (Product Delivery Intelligence & Optimization)

Jan 2025 – Present

- **AI measurement and insight.** Advanced AI measurement literacy across the ~4,000-person product delivery organization, teaching teams to distinguish AI adoption from AI impact (PR cycle time, review time, lead time for change, defect rate). Led a proof of concept measuring actual AI code percentage and analyzing AI sessions — both to guide model selection and to help individuals improve how they write requirements, steer the agent, scope work, and know when to stop.
- **Claude Code rollout.** Co-led the rollout: authored all user documentation, provided troubleshooting and support, and enabled Claude Code platform owners to upskill by sharing how GitHub Copilot is managed. Reached ~800+ weekly users within months and produced insight into where Copilot versus Claude Code excels.
- **DX platform rollout.** Prototyped, vetted via a 250-person POC, and rolled out DX (getdx.com) to ~4,000 product-delivery contributors, giving teams a data-driven way to surface their own wins and pain points and drive change through a five-step process (Snapshot, Triage, Review, Prioritize, Measure). Drove a cultural shift toward team-owned improvement, lifting all measured drivers versus onboarding baselines with a conservative ~\$10M annual ROI.

Sr. Software Engineer II · IS ExE EngEx (Engineer Experience)

Sep 2022 – Jan 2025

- **GitHub Copilot rollout.** Led it from prototype to company-wide rollout — a 250-person trial, business case to senior leadership, and partnership with legal, security, and risk for Principal's first AI tool. The trial proved 6%+ productivity gain against a 0.5% break-even; Copilot now serves ~3,500 weekly users, returning ~\$26M in annual productivity. Delivered training, demos, and a tech-conference session to 3,000+ engineers.
- **Engineering data pipeline.** Designed and built a pipeline feeding the company data lake (Snowflake) with curated data products, processing ~2M records/day synchronously and asynchronously at minimal overhead. Powers cloud security posture and misconfiguration alerts and finance billback of AI licensing.
- **Engineer workstation setup utility.** Built an engineer-first utility (Ansible; Windows and Linux via WSL) that reduced new-engineer setup from multiple business days to under an hour. Run by ~5,700 unique users in a year, saving ~91,000 engineer hours (~\$6.8M in recovered productivity).
- **License management modernization.** Cut excess MSDN and IBM RAD licensing for \$150K+ in annual savings and migrated the on-prem JetBrains license server to a cloud offering for scale and availability.

Sr. Software Engineer · IS CPE (Cloud Platform Engineering)

Apr 2021 – Sep 2022

- **Cloud landing zones and account vending.** Built and evolved Principal's AWS, GCP, and Azure cloud foundations, now supporting 3,500+ AWS accounts with new accounts vended in under an hour and foundational changes rolled out at scale. AWS became the preferred platform over on-prem alternatives.
- **IaC pre-deploy scanning.** Created a scanning solution (Checkov and custom checks), now required in Principal's production pipeline: ~2,700 runs/day across ~4,000 repositories, evaluating ~300 rules per run.

Prevented production regressions and materially improved cloud security posture.

- **Software delivery platforms via IaC.** Administered critical delivery platforms (GitHub, Azure DevOps) for 3,500+ engineers through infrastructure as code and automation, enabling cloud delivery while meeting regulatory and security controls.

Software Engineer · PGI Talent & Organizational Competency

Mar 2019 – Apr 2021

- **Technical interviewing via pair programming.** Assessed engineering candidates, improving new-hire quality and retention.
- **Lean engineering practice and consulting.** Coached teams through pairing, mobbing, and consulting to raise product quality.

Application Developer · PGI DESTination Program

May 2014 – Mar 2019

- **DESTination program future vision.** Key contributor to a program that redefined the business unit's technological future vision and the steps to reach it. Designed and delivered high-performing, high-impact systems in support of that vision.
- **Loosely coupled integration framework.** Designed and implemented a framework that lets systems be added, replaced, or removed with agility.

Earlier roles: Application Developer Intern, Principal Financial Group (2012–2013); Systems Administrator Assistant and Support Technician, Drake University (2011–2013).

SKILLS

AI & PRODUCTIVITY	AI-assisted development across the SDLC · AI tooling: evaluation, business case, administration & governance · AI enablement: training, documentation, pairing · AI impact measurement & analysis · GitHub Copilot · Claude Code · SPACE metrics · DORA metrics · DX (getdx.com) · Developer experience & productivity
PLATFORMS	Amazon Web Services (S3, SQS & SNS, Lambda, CloudWatch) · Google Cloud Platform (GCP) · Microsoft Azure · GitHub · Azure DevOps · Snowflake · PostgreSQL · Microsoft SQL Server · Mule ESB
LANGUAGES	Python · TypeScript · C# (MVC and Web API) · PowerShell · Java (limited) · Visual Basic.NET (limited)
INFRASTRUCTURE & TOOLING	Terraform · CloudFormation · AWS CDK · Ansible · Checkov · Windows Subsystem for Linux (WSL) · Workstation setup automation · Docker
PRACTICES	Platform engineering · Cloud security posture · Cost optimization & license management · Data pipelines · DevOps & CI/CD · Lean engineering · Technical leadership · Enablement & documentation

EDUCATION

Drake University

Des Moines, IA

Bachelor of Science in Business Administration

2014

- Information Systems and Marketing Major, Computer Science Minor
- Cumulative GPA 3.67/4.0 — Cum Laude, Departmental Honors

LEADERSHIP & RECOGNITION

- **Rebuilt trust with a chief accounting officer.** Led a team to deliver a product I defined — on time and defect-free — while training the engineers on AWS and industry-leading practices, rebuilding a chief accounting officer's trust in IT.
- **Praised for enterprise-wide improvement.** Repeatedly recognized by the director of engineering for improving the enterprise while addressing individual teams' needs.